

Pricing the Make-whole Bond

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Abstract

This paper presents a pricing system for Make-Whole bonds, a special type of callable bond whose redemption price is calculated based on a market benchmark interest rate plus a contractual spread, following the model of Lin Ma et al. (2023). The report details the technical implementation, including the strike price formula, data variable inventory, construction of future cash flows, and sample formation. Using bond data from the WRDS Mergent FISD database and the U.S. Treasury yield curve, the system analyzes 32,455 bonds with Make-Whole provisions, of which 20,684 are pure Make-Whole bonds. Results show that the strike price distribution is right-skewed and leptokurtic, reflecting the risk-compensation structure. Prices exhibit a strict negative correlation with Treasury rates, validating the pricing formula. The study also examines hybrid Make-Whole bonds, highlighting their two-phase behavior: early prices are highly sensitive to interest rates, while later prices converge toward a fixed call price. The system provides a fair, transparent, and market-responsive pricing tool that balances issuer flexibility with investor protection.

Keywords: Make-Whole Bond, Callable Bond Pricing, Strike Price Formula, Treasury Spread, Hybrid Bond, Interest Rate Risk

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1 Introduction

A Make-Whole Bond is a special type of callable bond whose redemption price is calculated based on a market benchmark interest rate plus a predetermined spread. The system is designed to implement the Make-Whole pricing model described in Lin Ma et al.(2023).

This report provides a detailed description of the technical implementation, algorithmic principles, result analysis, and improvement suggestions for the Make-Whole bond pricing system.

2 The Strike Price Formula

According to Lin Ma et al.(2023), the Make-Whole Call strike price for bond i on a potential call date t is calculated as:

$$P_{i,t}^{make-whole\ call} = \sum_{n=1}^N \frac{CF_{i,t+n}}{(1 + swap_rate_{i,t+n} + make_whole_premium_i)^n}$$

where:

- $CF_{i,t+n}$: Future cash flow (coupon and principal) at time $t + n$.
- $swap_rate_{i,t+n}$: Benchmark interest rate (swap rate) for maturity n at time t .
- $make_whole_premium_i$: Contractual spread added to the benchmark rate (the “make-whole” penalty).

At its core, the formula presented for the Make-Whole Call strike price, is a specific and sophisticated application of the fundamental principle of finance: the time value of money. This calculation determines the price a bond issuer must pay to redeem a bond early under a “make-whole call” provision, a common feature in modern corporate and municipal debt. Financially, its purpose is to disincentivize opportunistic refinancing by the issuer while compensating the bondholder fully for the loss of their investment. The issuer is allowed to call the bond, but must “make the investor whole” by paying a price that theoretically covers the present value of all remaining cash flows, discounted not at the bond’s original coupon rate, but at a current market benchmark rate plus a contractual penalty spread. This mechanism balances flexibility for the issuer with fair protection for the lender.

Mathematically, the formula is a sum of discounted cash flows, a standard valuation technique. The variable $CF_{i,t+n}$ represents the future cash flows (coupons and final principal) of bond i scheduled after the potential call date t . The index n counts the number of years from the call date t to each specific cash flow. The discount factor: $(1 + swap_rate_{i,t+n} + make_whole_premium_i)^n$. Here, $swap_rate_{i,t+n}$ is the benchmark interest rate, typically the swap rate for a maturity of n years observed at time t . The swap rate reflects the prevailing risk-free(or near risk-free) interest rate in the economy for that specific time horizon. Adding the make whole premium to this benchmark is the core of the make-whole mechanism. This premium is a fixed, contractual number (e.g., 25 basis points) agreed upon at bond issuance and acts as a punitive spread. The discount rate applied to

each cash flow is therefore the appropriate, maturity-matched benchmark rate plus this penalty. Using a maturity-matched rate (e.g., a 5-year cash flow is discounted by the 5-year swap rate plus the premium) is more accurate than using a single yield for all cash flows, as it properly accounts for the shape of the yield curve.

If an issuer wishes to call a bond because market interest rates have fallen, a traditional call provision at par would greatly disadvantage the holder, who must then reinvest the returned principal at lower prevailing rates. The make-whole call formula neutralizes this interest rate risk for the holder. By discounting the bond’s remaining cash flows at the current low market rates (the swap rates) plus a premium, the resulting present value will be higher than the bond’s standard market price. The lower the market rates, the lower the discount rates (swap rates), but the addition of the fixed premium mitigates this decrease slightly. Ultimately, the calculation produces a strike price that is generally at a premium to the bond’s face value and market price. This premium compensates the holder for the loss of their above-market coupon stream. Consequently, the make-whole call is rarely exercised purely for interest rate savings; it is typically invoked for structural reasons like mergers or asset sales. The formula thus elegantly translates a contractual protection into a precise, market-based mathematical price, ensuring the call price is fair, transparent, and responsive to changing financial conditions.

3 Data Variable Inventory

The bond data in this report is sourced from the WRDS **Mergent FISD** database, and the Treasury rate curve is sourced from the U.S. **Department of the Treasury**. The following tables list all relevant variables extracted from the provided data files. The “Role in Strike Price Construction” column explains how each variable is used in the model.

3.1 Bond Identifiers (Linking Keys)

These variables are used to uniquely identify bonds and link records across different tables.

Variable Name	Description	Role in Strike Price Construction
issue_id	A Mergent-generated number unique to each issue	Primary key for linking all bond-related information.
issuer_id	Issuer ID	Links to issuer-level information.
complete_cusip	Complete CUSIP	Unique security identifier for validation and external data linking.
issue_cusip	Issue CUSIP	Alternative security identifier.

3.2 Make-Whole Premium (Spread)

The contractual spread that forms the *make_whole_premium_i* component of the discount rate.

Variable Name	Description	Role in Strike Price Construction
make_whole_spread	The spread above a comparable treasury issue that will have to be paid if the make whole call option is exercised.	Direct source for $make_whole_premium_i$. This textual field (e.g., “0.50%”) must be parsed into a numeric value (e.g., 0.0050).
investment_grade	A flag indicating if the bond is investment grade (Y/N)	Used as a fallback rule: if make_whole_spread is missing, assume 25 bps for investment grade and 50 bps for high-yield bonds.

3.3 Bond Type Determination and Call Schedule

These variables determine whether the bond is a *pure* Make-Whole bond or a *hybrid* bond (with a fixed-price call following the Make-Whole period). This classification defines the endpoint $t + N$ and the final cash flow.

Variable Name	Description	Role in Strike Price Construction
make_whole_start_date	The date as of which a make whole call option becomes effective.	Defines the first possible call date t for the Make-Whole option.
make_whole_end_date	The date on which the make whole call option expires.	For a <i>pure</i> Make-Whole bond, this is the last possible call date. It helps define the period $[t, t + N]$.
next_call_date	The next date on which a call for redemption by the issuer would become effective.	For a <i>hybrid</i> bond, this is the date $t + N$ (the first fixed-price call date). Cash flows are calculated up to this date.
next_call_price	The call price (as % of par) associated with next_call_date .	For a <i>hybrid</i> bond, this is the final cash flow $CF_{i,t+N}$ (converted to dollar amount).
maturity	Maturity Date	For a <i>pure</i> Make-Whole bond, this is the date $t + N$. The final cash flow is the principal plus final coupon.
callable	A flag indicating that the issue is callable.	Used to filter bonds that are eligible for a Make-Whole call calculation.

3.4 Cash Flow (CF) Calculation

These variables are used to construct the schedule and amounts of future cash flows $CF_{i,t+n}$.

Variable Name	Description	Role in Strike Price Construction
principal_amt	Principal Amount (face value)	Used to calculate the dollar amount of coupon payments and the final principal repayment.
coupon	The current applicable annual interest rate (in %)	The annual coupon rate used to calculate periodic coupon payments.
interest_frequency	Code indicating how often interest payments will be made (e.g., SEMI-ANNUAL)	Determines the number of coupon payments per year and the timing of cash flows.
day_count_basis	Basis for determining interest paid during each period (e.g., 30/360)	Used for precise calculation of the accrued interest and coupon amount for each period.
dated_date	Date from which interest accrues	Reference point for calculating the first coupon period.
first_interest_date	Date of the first interest payment	Used to generate the schedule of coupon payment dates.
last_interest_date	Last interest payment date	Can be used to verify the coupon schedule near maturity.

4 Constructing Future Cash Flows (CF)

Constructing the future cash flow sequence (CF) for a bond requires generating all payment timings and amounts systematically from the potential call date t to the determined end date $t+N$ based on the bond's specific terms. The entire process begins with a critical determination of the bond type, which directly dictates the endpoint of the cash flows and the composition of the final payment. The judgment is primarily derived from bond redemption clause data: for a pure "make-whole" bond, its "make-whole" end date is essentially consistent with the bond's maturity date, and there is no earlier fixed-price call date, so the cash flows will continue until the legal maturity date of the bond, with the final payment being the principal plus the last coupon; conversely, for a "hybrid" bond, its "make-whole" provision ends before the maturity date and gives way to a definite fixed-price call date, at which point the endpoint of the cash flows is that fixed-price call date, and the final payment is the contractually specified fixed call price (usually expressed as a percentage of par) multiplied by the principal amount, rather than the principal at maturity.

After determining the time frame for the cash flows, the next step is to identify all coupon payment dates within this period. This requires utilizing basic bond information, including the

annual coupon rate, payment frequency, first interest date, and maturity date, and following the specified day count convention. Starting from the first interest date, all coupon payment points from the call date t to the end date $t + N$ can be derived based on the payment frequency (e.g., semi-annually). Next, calculate the specific cash flow amount at each point. For each coupon payment during the period, the amount is determined by multiplying the principal by the coupon rate, and then adjusting for the payment frequency and the actual day count fraction for the accrual period (according to the day count convention). For the final payment, based on the bond type determined in the first step, the corresponding calculation rule is applied: if it is a pure “make-whole” bond, it is the principal plus the last coupon; if it is a “hybrid” bond, it is the pre-agreed fixed call price. Finally, the series of future time points $t + n$ and their corresponding precise cash payment amounts $CF_{i,t+n}$ are paired and combined, forming the complete cash flow input sequence for calculating the “make-whole” strike price.

Bond Type	Judgment Criteria (Fields from REDEMPTION table)	Cash Flow Endpoint ($t + N$)	Final Cash Flow
Pure Make-Whole Bond	<code>make_whole_end_date</code> $\approx$ maturity and <code>next_call_date</code> is NULL or later than maturity.	Bond Maturity Date (maturity)	<code>principal_amt</code> + Final Coupon
Hybrid Bond	<code>make_whole_end_date</code> < maturity and <code>next_call_date</code> has a value (first fixed-price call date).	First Fixed-Price Call Date (<code>next_call_date</code>)	Fixed Call Price: <code>next_call_price</code> (as % of par) $\times$ <code>principal_amt</code> / 100

5 Sample Construction

This report selects all bonds with a makewhole call provision from the **Mergent FISD** database, amounting to 32,455 bonds. The mean coupon rate is 5.37%, with a standard deviation of 3.11, indicating moderate dispersion around the average. The median coupon is 5.15%, which is slightly lower than the mean, suggesting a mild right-skew in the distribution. This is confirmed by the skewness statistic of 6.22 and the very high kurtosis value of 142.14, which together indicate a strongly right-skewed and leptokurtic distribution—meaning the data has a heavy tail on the higher end and is more peaked than a normal distribution. The range of coupon rates is notably wide, from a minimum of 0% to a maximum of 99.876%. The latter appears to be an extreme outlier, which likely contributes to the high skewness and kurtosis. The central 50% of bonds (between the 25th and 75th percentiles) have coupon rates ranging from 3.75% to 6.625%, reflecting a concentration of bonds with moderate coupon levels. Notably, 90% of bonds have coupon rates at or below 8.625%, while only 1% exceed 12%.

The make-whole spread has 24,379 observations from the database. Its mean value is approximately 33.24 basis points, with a standard deviation of 15.81, indicating moderate variation around the mean. The distribution is positively skewed (skewness = 0.73), meaning it has a longer right tail. This is confirmed by the percentiles: while the median (50th percentile) is 30 bps, the mean is higher at 33.24 bps. The distribution is also highly leptokurtic (kurtosis = 12.32). The majority

of spreads are concentrated between 20 bps (25th percentile) and 50 bps (75th percentile) and 90% of the observations have a spread of 50 bps or less. There are extreme values on the high end, with the maximum observed value reaching 375 bps.

For missing values, this report replace them with 50 bps (Median).

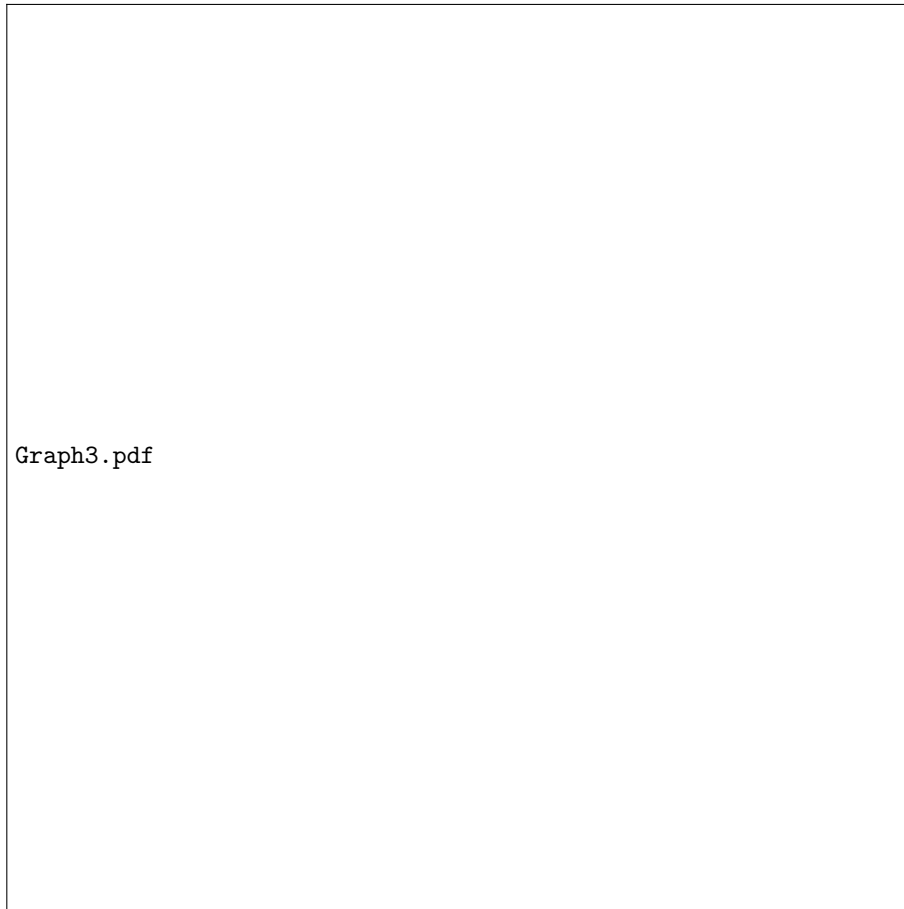


Figure 1

6 Calculation Results for Pure Make-Whole bonds

There are 20,684 observes in the category of pure make-whole bond. The pricing results are shown in Table 5. We first calculated the strike price of each pure make-whole bond at the next callable date.

Table 5: Descriptive Statistics for price_per_100

Percentiles		Extreme Values	
Percentile	Value	Type	Value
1%	95.11591	Smallest	56.5564
5%	99.68797		70.51692
10%	100.8135		71.07095
25%	104.7414		72.0565
50%	114.0078		
75%	127.5653	Largest	220.2088
90%	139.4264		222.9951
95%	146.3837		226.7954
99%	162.6322		234.8361
Summary Statistics			
Statistic		Value	
Obs		20,684	
Sum of wgt.		20,684	
Mean		117.547	
Std. dev.		15.76789	
Variance		248.6263	
Skewness		1.047201	
Kurtosis		4.678453	

From a quantile perspective, the prices are mainly concentrated in the range of 100 to 146 (the 10th percentile is 100.81, and the 95th percentile is 146.38). 50% of the observations fall below 114.0078 (median), and 75% are below 127.5653. Only 1% of bond prices exceed 162.63. However, the presence of extreme values further confirms the heavy-tailed characteristic: the four smallest values (56.5564 to 72.0565) are significantly lower than the 1st percentile (95.11591), while the four largest values (220.2088 to 234.8361) are substantially higher than the 99th percentile (162.6322).

Figure 3 shows the price distribution of the strike price for make-whole bonds.

The occurrence of such extreme values is reasonable. Based on the characteristics of callable bonds, we know that the call date has a significant impact on the call price. Taking a bond from the sample as an example (CUSIP: 65504L), its coupon rate is 8.95%, and on the calculation date, the U.S. market interest rate was at a low point (2.67%), resulting in a large annualized spread. Moreover, its 30-year long term significantly amplifies the enhancing effect of this spread on the present value when calculating the price. From this, we can draw some conclusions: the timing of the call has a huge impact on bond prices, and interest rates affect the call price, causing price fluctuations. This mechanism aims to protect investors: when market interest rates fall, although issuers may tend to call high-coupon bonds to reduce financing costs, the Make-Whole provision uses the Treasury rate and a premium as the discount benchmark to ensure that the call price covers the reinvestment risk faced by investors due to the early termination of the bond, meaning investors can obtain returns similar to holding the bond to maturity. Therefore, in the current low-interest-rate environment, even if the bond trades at a significant premium, the price still strictly follows the Make-Whole pricing formula, reflecting the protective nature of the provision for investors' interests. The calculation results are consistent with financial logic and the intent of the contract design.

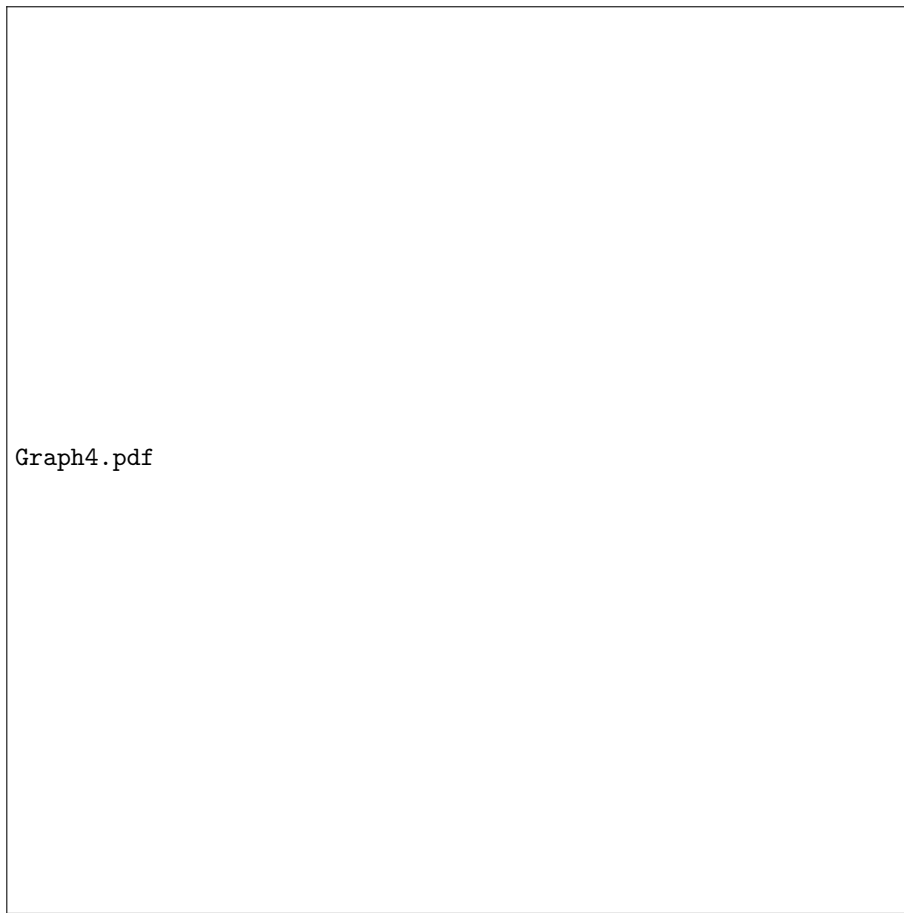


Figure 2

6.1 Make-Whole Treasury Spread Distribution (Pure)

We define the Make-Whole Treasury Spread to represent the excess return from bond redemption, in order to analyze the bond's risk and compensatory measures. Table 6 shows the descriptive statistics for the Make-Whole Treasury Spread.

Table 6: Descriptive Statistics for the Make-Whole Treasury Spread

Distribution and Extremes		Summary Statistics	
Percentile	Value	Statistic	Value
1%	-0.00537	Obs (<i>n</i>)	20,685
5%	0.00394	Sum of Weights	20,685
10%	0.00641	Mean	0.03020
25%	0.01197	Std. Dev.	0.02432
50% (Median)	0.02320	Variance	0.00059
75%	0.04318	Skewness	1.12939
90%	0.06633	Kurtosis	4.04471
95%	0.07972		
99%	0.10201		
Extreme Values			
Smallest	Largest		
-0.03800	0.14061		
-0.03623	0.14805		
-0.03126	0.15103		
-0.03100	0.17316		

Note: The spread is calculated as *Coupon Rate* - (*Weighted Average Treasury Rate* + *MW Premium*). The distribution exhibits right skewness (Skewness > 0) and leptokurtosis (Kurtosis > 3).

Graph6.pdf

Figure 3

From a mathematical perspective, the distribution of this spread exhibits typical non-normal characteristics. Its mean (0.03020) is higher than the median (0.02320), and combined with a positive skewness (1.12939), this mathematically confirms a right-skewed distribution. This indicates that the spreads of most bonds are concentrated in the region below the mean, but there are a few bonds with very high spreads, pulling the overall average to the right. Meanwhile, the kurtosis (4.04471) is significantly greater than 3, indicating that the distribution has leptokurtic (fat-tailed) characteristics, meaning the data is more concentrated around the mean, but the probability of extreme values is higher than in a normal distribution. This right-skewed, fat-tailed shape is a common feature of financial return data. Percentile data (e.g., the 1st percentile is negative, and the 99th percentile reaches 0.10201) and extreme values (minimum -0.03800, maximum 0.17316) together outline the full range and dispersion of the data, while the standard deviation (0.02432) quantifies the volatility of the spread around the mean.

These statistics reveal the risk structure and pricing logic of the Make-Whole bond market. The essence of the spread calculation formula, $\text{Spread} = \text{Coupon Rate} - (\text{Treasury Rate} + \text{MW Premium})$, is the excess return provided by the bond to investors, compensating for credit risk, liquidity risk, and potential early redemption risk. The right-skewed distribution indicates that while the mainstream market (around the median) offers relatively moderate excess returns, there exists a "high-risk, high-return" tail. This tail likely corresponds to lower-rated high-yield bonds, whose higher coupon rates pull up the overall spread level.

Specifically, bonds with negative spreads (such as the minimum extreme value) imply that their coupon rates are even lower than the risk-free benchmark. For these bonds, the Make-Whole provision is in a state of "deep out-of-the-money," meaning the redemption protection is temporarily ineffective. Conversely, bonds with high spreads (such as those at the 99th percentile) signify that investors demand and receive substantial risk premiums.

7 Make-Whole Bond Strike Price Curves

Due to limited access to the database, we are unable to retrieve the complete call schedule for each bond. Therefore, we have chosen to calculate the strike prices for the entire period from the make-whole start date to the make-whole end date, using a monthly interval as the step size.

We have developed a program capable of calculating the strike price curve for every bond containing a make-whole call provision. These curves reveal how the price of a make-whole bond is influenced by fluctuations in Treasury rates. As is plotted in figure 4, the curves of the prices and treasury rates have shown some interesting facts to us.

Note: figure 4 and figure 6 shows the curve of Bond ISSUE ID: 103417 CUSIP: 271790

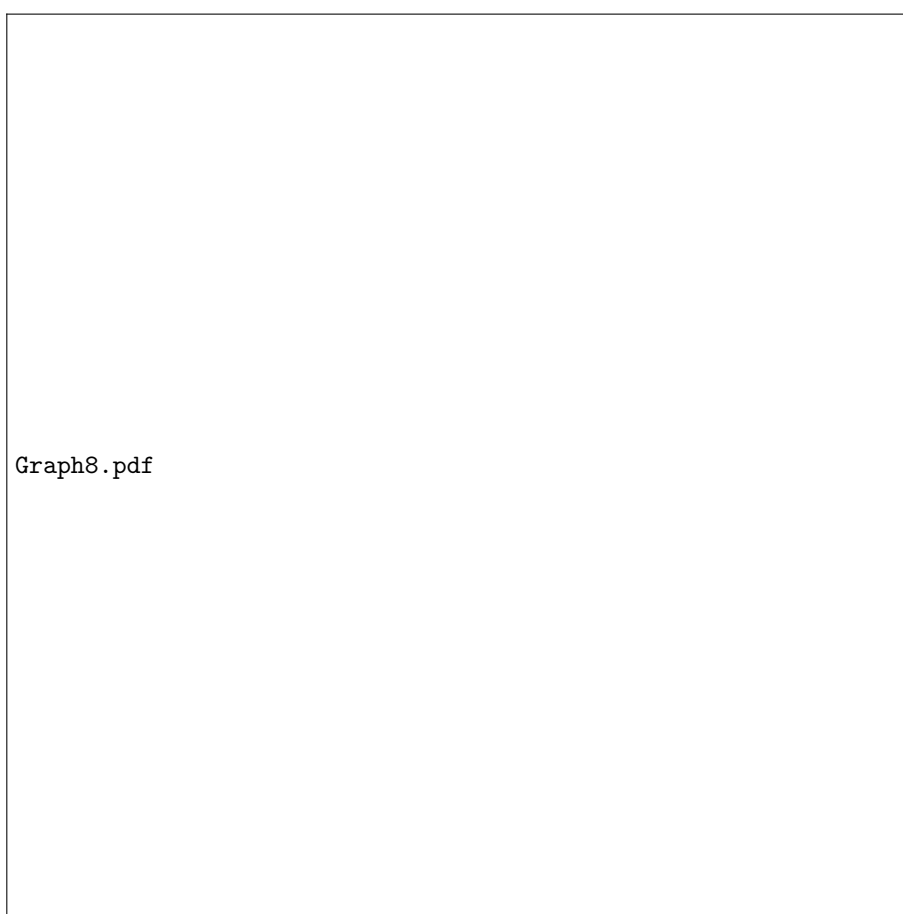


Figure 4

Bond prices exhibit a strict negative correlation with treasury rates, forming an almost perfect inverse mirror image throughout the sample period. This directly stems from their pricing formula—future cash flows are discounted at the "Treasury yield plus a fixed premium." Consequently, fluctuations in interest rates are directly and inversely transmitted to prices.

From a historical perspective, the curve can be clearly divided into three distinct phases, each corresponding to a different dominant logic.

During the interest rate cutting cycle from 2002 to 2004, the decline in Treasury yields directly pushed Make-Whole bond prices to historical highs, perfectly illustrating the "duration effect" of bonds benefiting from falling interest rates.

The rate-hiking cycle from 2004 to 2007 revealed the opposite side: prices trended downward as interest rates rose.

The most revealing phase occurred after 2007: despite the sharp drop in interest rates following the 2008 financial crisis, prices failed to reach new highs. Instead, they continued to decline in the low-interest-rate environment until maturity. This highlights the "dual nature" and dynamic shift of factors influencing prices. In the early stages of a bond's life, interest rate risk dominates price fluctuations. As maturity approaches, the time decay effect—where the present value of future cash flows naturally diminishes as time shortens—becomes the stronger driving force. This compels prices to converge toward par value, regardless of earlier fluctuations.

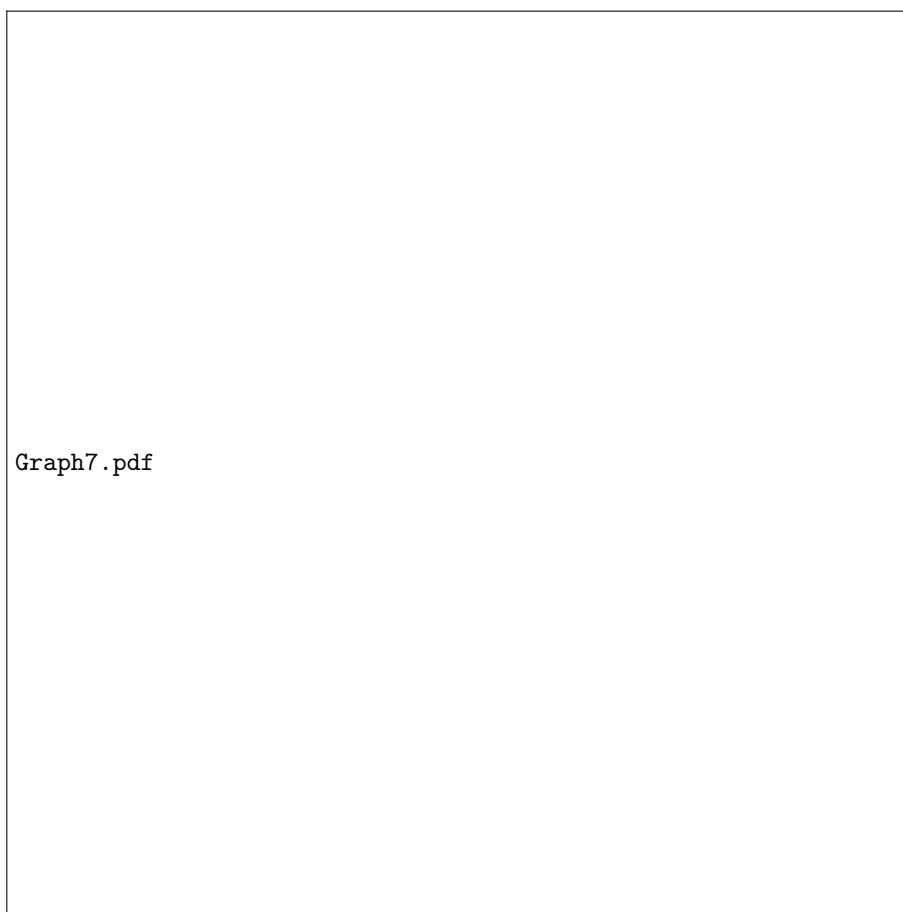


Figure 5

8 Hybrid Bond pricing

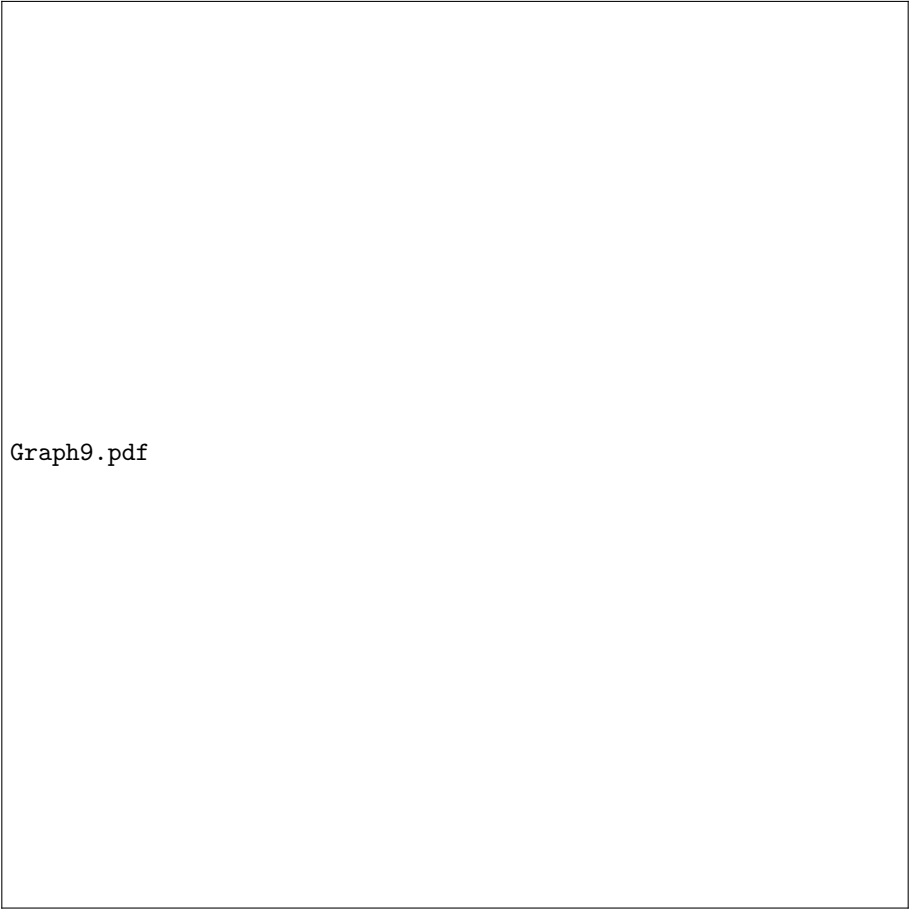
Due to the lack of call schedule data, though our program has the capability to calculate hybrid bond prices, we are currently unable to obtain valid results.

The design of the call provisions for hybrid Make-Whole bonds reflects the sophisticated trade-offs in financial engineering aimed at balancing the interests of issuers and investors. These bonds combine two redemption mechanisms: during the early call period, the issuer can redeem the bonds at a Make-Whole price based on market interest rates plus a fixed premium; after a specified date, redemption can occur at a predetermined fixed price (typically declining over time). This dual

structure provides issuers with significant financing flexibility—when market interest rates fall substantially, they can redeem high-coupon bonds at a cost close to fair value via the Make-Whole provision; when interest rates are relatively stable, the fixed-price redemption offers a known cost ceiling, aiding cash flow planning and debt management.

For investors, this hybrid structure introduces dual uncertainty: not only is the timing of redemption difficult to predict, but the redemption method also varies with market conditions, significantly increasing uncertainty regarding the actual holding period and final yield. Consequently, investors typically demand an additional yield premium (research suggests approximately 10–50 basis points) as compensation, reflecting the value of the embedded option and compensation for reinvestment risk.

From a pricing model perspective, the calculation of the Make-Whole price for hybrid bonds is limited to the period before the first fixed-price call date, and includes the fixed call price rather than the principal at that date. This represents a key difference from pure Make-Whole bonds, which are discounted over the full maturity period. This structure results in a distinct phased behavior in bond pricing: in the early stage, pricing is primarily driven by the Make-Whole provision, making it highly sensitive to interest rate changes; in the middle stage, prices fluctuate between the Make-Whole value and the fixed call price; as the fixed-price call date approaches, prices gradually converge toward the call price, effectively forming a "price ceiling." This price path further influences the bond's interest rate risk profile—as the call date nears, the sensitivity of the price to interest rate changes diminishes.



Graph9.pdf

Figure 6